

String Problems

C program to check whether a string is palindrome or not?

C program to check whether a string is palindrome or not?

```
#include <stdio.h>
#include <string.h>

int main() {
    char string1[20];
    int i, length;
    int flag = 0;

    // Prompt the user for input
    printf("Enter a string: ");
    scanf("%s", string1);

    // Calculate the string length
    length = strlen(string1);

    // Compare characters from the start and end of the string
    // and stop if a mismatch is found or the middle of the string is reached.
    for (i = 0; i < length / 2; i++) {
        if (string1[i] != string1[length - i - 1]) {
            flag = 1;
            break;
        }
    }

    // Output the result
    if (flag) {
        printf("%s is not a palindrome\n", string1);
    } else {
        printf("%s is a palindrome\n", string1);
    }

    return 0;
}
```

C program to reverse the order of words in a string

```
/**
 * C program to reverse order of words in a string
 */
#include <stdio.h>
#include <string.h>
#define MAX_SIZE 100 // Maximum string size

int main()
{
    char str[100], reverse[100];
    int len, i, index, wordStart, wordEnd;

    printf("Enter any string: ");
    gets(str);

    len = strlen(str);
    index = 0;

    // Start checking of words from the end of string
    wordStart = len - 1;
    wordEnd = len - 1;

    while(wordStart > 0)
    {
        // If a word is found
        if(str[wordStart] == ' ')
        {
            // Add the word to the reverse string
            i = wordStart + 1;
            while(i <= wordEnd)
            {
                reverse[index] = str[i];

                i++;
                index++;
            }
            reverse[index++] = ' ';

            wordEnd = wordStart - 1;
        }

        wordStart--;
    }

    // Finally add the last word
    for(i=0; i<=wordEnd; i++)
    {
        reverse[index] = str[i];
        index++;
    }

    // Add NULL character at the end of reverse string
    reverse[index] = '\0';

    printf("Original string\n%s\n", str);
    printf("Reverse ordered words\n%s", reverse);

    return 0;
}
```